Logic at
Joy of Problem Solving 2026

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Abstract

These are notes for a part of the Logic Module taught at the ACM India Winter School 2026, *Joy of Problem Solving*, hosted at IIT Madras.

Chapter 1 is a pre-reading on the syntax and semantics of propositional logic, to be read before the first lecture.

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Propositional Logic

"Not the same thing a bit!" said the Hatter. "You might just as well say that 'I see what I eat' is the same thing as 'I eat what I see'!"
Lewis Carroll, *Alice's Adventures in Wonderland*

1.1 Propositions

A *proposition* is a statement that is either true or false. "91 is prime" is a proposition (a false one, since $91 = 7 \times 13$); "Close the door" and "$x > 3$" are not. Propositions combine into larger ones using words such as *not*, *or*, *and* and *if ... then*, and the truth value of a combination depends only on the truth values of its parts. Propositional logic studies what can be concluded from this structure alone. George Boole treated propositions algebraically in 1847 and 1854.[1]

[1] G. Boole, *The Mathematical Analysis of Logic*, Macmillan, Barclay and Macmillan, Cambridge, 1847; G. Boole, *An Investigation of the Laws of Thought*, Walton and Maberly, London, 1854. Boole wrote xy for "x and y" and worked with the equation $x^2 = x$, which holds for the truth values 0 and 1; Boolean algebra is named after him.

A puzzle On an island every inhabitant is either a *knight*, who always tells the truth, or a *knave*, who always lies.[2] Of two inhabitants, A says "B is a knave" and B says "A and I are of the same kind". Which of them is a knight? The answer is worked out in Section 1.4.

1.2 Syntax

Fix a nonempty, finite or countably infinite set of *atomic propositions* (*atoms* for short), listed without repetitions as $\mathcal{P} = \{p_1, p_2, \dots\}$ and also written p, q, r, and two *logical connectives*, $\neg$ (read as *not*) and $\vee$ (read as *or*).[3] Formulas are strings over $\mathcal{P} \cup \{\neg, \vee, (,)\}$, where each atomic proposition counts as a single symbol, different from $\neg$, $\vee$, (and).

Definition 1.2.1 The set Φ of *formulas* is the smallest set of strings satisfying the following conditions, that is, the intersection of all sets of strings that satisfy them.

- Every atomic proposition p is a member of Φ.
- If α is a member of Φ, so is $(\neg\alpha)$.
- If α and β are members of Φ, so is $(\alpha \vee \beta)$.

The Greek letters α, β, γ stand for arbitrary formulas; they are not symbols of the language. For example, $((\neg(p \vee q)) \vee (\neg r))$ is a formula, built from p, q, r by four applications of the rules. We omit the outermost parentheses, and $\neg$ binds more tightly than $\vee$, so this formula is written $\neg(p \vee q) \vee \neg r$.

Structural induction principle *Smallest* means that Φ is a subset of every set satisfying the three conditions. This gives a method of proof: let S be a set such that every atomic proposition is in S, $(\neg\alpha) \in S$ whenever $\alpha \in S$, and $(\alpha \vee \beta) \in S$ whenever $\alpha, \beta \in S$. Then $\Phi \subseteq S$.

To prove that every formula has a property, we take S to be the set of strings with that property.

Unique readability Because of the parentheses in the rules, each formula is built in only one way. Without them, the string $\neg p \vee q$ could be built either as the negation of $p \vee q$ or as the disjunction of $\neg p$ and q.

Proposition 1.2.2 (Unique readability) *Every formula is of exactly one of the forms p, $(\neg\alpha)$ and $(\alpha \vee \beta)$, and the formulas α and β are uniquely determined.*

We omit the proof of unique readability.

Hence each formula can be drawn, in exactly one way, as a *parse tree* with atomic propositions at the leaves and connectives inside; Figure 1.1 shows an example.

[2] Puzzles of this kind were made popular by Raymond Smullyan, starting with R. M. Smullyan, *What Is the Name of This Book? The Riddle of Dracula and Other Logical Puzzles*, Prentice-Hall, Englewood Cliffs, 1978.

[3] The notation of these notes follows M. Mukund and S. P. Suresh, *An Introduction to Logic*, lecture notes, Chennai Mathematical Institute, 2011.

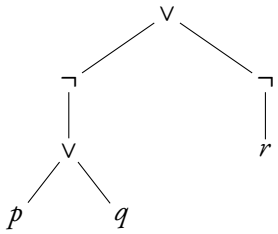


Figure 1.1 The parse tree of $\neg(p \vee q) \vee \neg r$, that is, of $((\neg(p \vee q)) \vee (\neg r))$.

Unique readability also lets us define functions on formulas by recursion: if we give a value for each atomic proposition and say how the values on $\neg\beta$ and $\beta \vee \gamma$ are computed from those on β and γ, this determines a unique value for each formula. For example, the *vocabulary* $Voc(\alpha)$, the set of atomic propositions in α, is given by $Voc(p) = \{p\}$, $Voc(\neg\beta) = Voc(\beta)$ and $Voc(\beta \vee \gamma) = Voc(\beta) \cup Voc(\gamma)$. By structural induction, $Voc(\alpha)$ is always finite.

Derived connectives

$$\alpha \wedge \beta \stackrel{\text{def}}{=} \neg(\neg\alpha \vee \neg\beta)$$

$$\alpha \supset \beta \stackrel{\text{def}}{=} \neg\alpha \vee \beta$$

$$\alpha \equiv \beta \stackrel{\text{def}}{=} (\alpha \supset \beta) \wedge (\beta \supset \alpha)$$

The connective $\wedge$ is read as *and*, $\supset$ as *implies*[4] and $\equiv$ as *if and only if*. Since these are abbreviations, proofs by structural induction still need only the cases p, $\neg\alpha$ and $\alpha \vee \beta$. The connectives bind in the order $\neg$, $\wedge$, $\vee$, $\supset$, $\equiv$, from tightest to loosest; $\wedge$ and $\vee$ group to the left, $\supset$ groups to the right, and repeated $\equiv$ is always parenthesised.

1.3 Semantics

Let $\top$ denote the truth value true and $\bot$ the truth value false.

Definition 1.3.1 A *valuation* (or *interpretation*) is a function $v : \mathscr{P} \to \{\top, \bot\}$.

We extend each valuation v to a map $\hat{v} : \Phi \to \{\top, \bot\}$ by recursion.

- For $p \in \mathscr{P}$, $\hat{v}(p) = v(p)$.
- $\hat{v}(\neg\beta) = \top$ if $\hat{v}(\beta) = \bot$, and $\bot$ otherwise.
- $\hat{v}(\beta \vee \gamma) = \bot$ if $\hat{v}(\beta) = \hat{v}(\gamma) = \bot$, and $\top$ otherwise.

By unique readability, $\hat{v}$ is well defined. We write v for $\hat{v}$, and $v \models \alpha$ ("v satisfies α") when $v(\alpha) = \top$.

Proposition 1.3.2 *For every valuation v, the connectives take the following values.*

[4] Giuseppe Peano wrote a reversed letter C for "implies" in *Arithmetices principia, nova methodo exposita* (1889), and Whitehead and Russell used $\supset$ in *Principia Mathematica* (1910–1913). Many books now write $\to$ instead.

$v(\alpha)$	$v(\beta)$	$v(\neg\alpha)$	$v(\alpha \vee \beta)$	$v(\alpha \wedge \beta)$	$v(\alpha \supset \beta)$	$v(\alpha \equiv \beta)$
$\top$	$\top$	$\bot$	$\top$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\top$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\top$	$\bot$	$\top$	$\bot$
$\bot$	$\bot$	$\top$	$\bot$	$\bot$	$\top$	$\top$

Proof. For $\wedge$: $v(\neg(\neg\alpha \vee \neg\beta)) = \top$ if and only if $v(\neg\alpha) = v(\neg\beta) = \bot$, that is, $v(\alpha) = v(\beta) = \top$. For $\supset$: $v(\neg\alpha \vee \beta) = \bot$ if and only if $v(\alpha) = \top$ and $v(\beta) = \bot$. The column for $\equiv$ follows from these two.

The disjunction $\vee$ is inclusive: $\alpha \vee \beta$ is true when both parts are. The implication $\alpha \supset \beta$ is false only when α is true and β false. For example, "if you score above 90, you get an A" is false only for a student who scores above 90 and does not get an A. So $\alpha \supset \beta$ is true whenever α is false.

Proposition 1.3.3 *If valuations v_1 and v_2 agree on $Voc(\alpha)$, then $v_1(\alpha) = v_2(\alpha)$.*

Proof. By structural induction on α. For $\alpha = p$ this is the assumption. For $\alpha = \neg\beta$, $Voc(\beta) = Voc(\alpha)$, so $v_1(\beta) = v_2(\beta)$ and hence $v_1(\alpha) = v_2(\alpha)$. For $\alpha = \beta \vee \gamma$, v_1 and v_2 agree on $Voc(\beta)$ and on $Voc(\gamma)$, so $v_1(\beta) = v_2(\beta)$ and $v_1(\gamma) = v_2(\gamma)$, and hence $v_1(\alpha) = v_2(\alpha)$.

If $Voc(\alpha)$ has n elements, there are 2^n ways to give them truth values, and by Proposition 1.3.3 these determine $v(\alpha)$ for every v. A *truth table* lists the 2^n cases, one per row. Truth tables in this form were introduced in 1921, independently, by Emil Post and Ludwig Wittgenstein.[5] The following truth table for $((p \supset q) \wedge p) \supset q$ has a column for each subformula.

p	q	$p \supset q$	$(p \supset q) \wedge p$	$((p \supset q) \wedge p) \supset q$
$\top$	$\top$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\top$	$\bot$	$\top$
$\bot$	$\bot$	$\top$	$\bot$	$\top$

1.4 Satisfiability and consequence

A formula α is *satisfiable* if $v \vDash \alpha$ for some valuation v. For example, $p \supset q$ is satisfiable and $p \wedge \neg p$ is not. By Proposition 1.3.3, a truth table decides whether a formula is satisfiable: it is satisfiable if and only if some row gives $\top$.

[5] E. L. Post, "Introduction to a general theory of elementary propositions", *American Journal of Mathematics* 43 (1921), 163–185; L. Wittgenstein, "Logisch-philosophische Abhandlung", *Annalen der Naturphilosophie* 14 (1921), 185–262, translated as *Tractatus Logico-Philosophicus*, 1922. Post also proved that every formula true under all valuations can be derived in the proof system of Whitehead and Russell's *Principia Mathematica*: the first published completeness theorem. Paul Bernays had proved the same result in 1918, in an unpublished Habilitationsschrift.

Consequence Let X be a set of formulas. A formula α is a *logical consequence* of X, written $X \models \alpha$, if every valuation that satisfies all formulas in X satisfies α. The truth table for $((p \supset q) \wedge p) \supset q$ in Section 1.3 shows $\{p \supset q,\ p\} \models q$, while $\{p \supset q,\ q\} \not\models p$.

***Exercise** 1.4.1* Let $n \geq 1$. Prove that $\{\alpha_1, \ldots, \alpha_n\} \models \beta$ if and only if $\alpha_1 \wedge \ldots \wedge \alpha_n \wedge \neg\beta$ is not satisfiable.

The puzzle, solved Let a stand for "A is a knight" and b for "B is a knight". A speaker is a knight exactly when what they say is true, so the two statements become $a \equiv \neg b$ and $b \equiv (a \equiv b)$. A solution is a valuation satisfying both.

a	b	$a \equiv \neg b$	$a \equiv b$	$b \equiv (a \equiv b)$
$\top$	$\top$	$\bot$	$\top$	$\top$
$\top$	$\bot$	$\top$	$\bot$	$\top$
$\bot$	$\top$	$\top$	$\bot$	$\bot$
$\bot$	$\bot$	$\bot$	$\top$	$\bot$

Only the second row satisfies both, so A is a knight and B is a knave, and this is the only solution.

***Exercise** 1.4.2* Two other inhabitants of the island are A and B. A says: "At least one of us is a knave." What are A and B?

***Exercise** 1.4.3* Exactly one of Asha, Bilal and Chitra broke the window. Asha says "Bilal did it." Bilal says "I did not." Chitra says "I did not." Exactly one of the three statements is true. Write these facts as formulas and find who broke the window. (Let a, b, c stand for "Asha broke it", "Bilal broke it", "Chitra broke it". "Exactly one of α, β, γ" is $(\alpha \wedge \neg\beta \wedge \neg\gamma) \vee (\neg\alpha \wedge \beta \wedge \neg\gamma) \vee (\neg\alpha \wedge \neg\beta \wedge \gamma)$.)

Compactness

Mathematics is the science of the infinite, its goal the symbolic comprehension of the infinite with human, that is finite, means.
Hermann Weyl, *The Open World*

2.1 Satisfiable sets of formulas

Chapter 1 asked whether a single formula is satisfiable. Here we ask the same question of a set of formulas, which may be infinite.

Definition 2.1.1 Let $X \subseteq \Phi$. A valuation v *satisfies* X, written $v \vDash X$, if $v \vDash \beta$ for every $\beta \in X$. The set X is *satisfiable* if some valuation satisfies it. We write $Y \subseteq_{\text{fin}} X$ when Y is a finite subset of X, and call X *finitely satisfiable* if every $Y \subseteq_{\text{fin}} X$ is satisfiable.

A finite nonempty set $X = \{\beta_1, \dots, \beta_n\}$ is satisfiable if and only if the single formula $\beta_1 \wedge \dots \wedge \beta_n$ is, so a truth table decides the question. An infinite set may mention infinitely many atomic propositions, and then no truth table is available. Every finite subset of X may be satisfiable, each by its own valuation, and it is not clear that these valuations can be combined into one. The compactness theorem shows that they can.

Theorem 2.1.2 (Compactness) *Let $X \subseteq \Phi$. Then X is satisfiable if and only if it is finitely satisfiable.*

If $v \vDash X$ and $Y \subseteq X$, then $v \vDash \beta$ for every $\beta \in Y$, so $v \vDash Y$. For the converse we extend a finitely satisfiable set as far as possible, and read a valuation off the result.

Kurt Gödel proved compactness in 1930, for countable languages, as a consequence of his completeness theorem for first-order logic; and the propositional version is a special case.[1] Anatoly Mal'cev proved it for languages of any size in 1936, and later used it to prove theorems about infinite groups.[2] The proof below works directly with satisfiability and needs no proof system.

***Exercise** 2.1.3* Let $\mathscr{P}$ be infinite and $X = \{p_1\} \cup \{p_i \supset p_{i+1} : i \geq 1\}$. Show that X is satisfiable, that $X \cup \{\neg p_5\}$ is not, and find a finite subset of $X \cup \{\neg p_5\}$ that is not satisfiable.

2.2 *Maximal finitely satisfiable sets*

Definition 2.2.1 A finitely satisfiable set $\Delta \subseteq \Phi$ is *maximal* if for every formula $\alpha \notin \Delta$, the set $\Delta \cup \{\alpha\}$ is not finitely satisfiable.

Example 2.2.2 For a valuation v, the set $\Delta_v = \{\alpha : v \vDash \alpha\}$ is maximal finitely satisfiable. The valuation v satisfies it. If $\alpha \notin \Delta_v$, then $v \vDash \neg\alpha$, so $\neg\alpha \in \Delta_v$, and the finite set $\{\alpha, \neg\alpha\}$ shows that $\Delta_v \cup \{\alpha\}$ is not finitely satisfiable.

The proof of compactness reverses this, and builds a valuation from a maximal finitely satisfiable set.

Lemma 2.2.3 *If Z is finitely satisfiable and α is a formula, then $Z \cup \{\alpha\}$ or $Z \cup \{\neg\alpha\}$ is finitely satisfiable.*

Proof. Suppose neither is. Then there are finite $Y_1, Y_2 \subseteq Z$ such that $Y_1 \cup \{\alpha\}$ and $Y_2 \cup \{\neg\alpha\}$ are not satisfiable. The finite set $Y_1 \cup Y_2 \subseteq Z$ is satisfied by some valuation v. If $v(\alpha) = \top$, then $v \vDash Y_1 \cup \{\alpha\}$; if $v(\alpha) = \bot$, then $v \vDash Y_2 \cup \{\neg\alpha\}$. Either way we have a contradiction.

The next lemma is due to Adolf Lindenbaum, a Polish logician killed in 1941, during the Second World War; it was first published by Alfred Tarski.[3]

Lemma 2.2.4 (Lindenbaum) *Every finitely satisfiable set X is contained in a maximal finitely satisfiable set.*

[1] K. Gödel, "Die Vollständigkeit der Axiome des logischen Funktionenkalküls", *Monatshefte für Mathematik und Physik* 37 (1930), 349–360. A proof uses only finitely many premises, so if every finite subset of X is consistent, then so is X; completeness turns consistency into satisfiability.

[2] A. Mal'cev (published as A. Malcev), "Untersuchungen aus dem Gebiete der mathematischen Logik", *Matematicheskii Sbornik* 1(43) (1936), 323–336.

[3] A. Tarski, "Über einige fundamentale Begriffe der Metamathematik", *Comptes Rendus des Séances de la Société des Sciences et des Lettres de Varsovie, Classe III* 23 (1930), 22–29; a fuller account is in A. Tarski, "Fundamentale Begriffe der Methodologie der deduktiven Wissenschaften. I", *Monatshefte für Mathematik und Physik* 37 (1930), 361–404. Tarski stated it for consistent sets of sentences in a proof system; the version here uses finite satisfiability instead.

Proof. The set Φ is countable: its alphabet $\mathscr{P} \cup \{\neg, \vee, (,)\}$ is countable, and for each length k there are countably many strings of length k. List the formulas as $\alpha_1, \alpha_2, \ldots$. Let $\Delta_0 = X$, and let

$$\Delta_{n+1} = \begin{cases} \Delta_n \cup \{\alpha_{n+1}\} & \text{if this set is finitely satisfiable,} \\ \Delta_n \cup \{\neg\alpha_{n+1}\} & \text{otherwise.} \end{cases}$$

By Lemma 2.2.3 and induction on n, every Δ_n is finitely satisfiable. Let $\Delta = \bigcup_n \Delta_n$ (Figure 2.1). A finite $Y \subseteq \Delta$ lies in some Δ_n, since $\Delta_0 \subseteq \Delta_1 \subseteq \cdots$, so Y is satisfiable; hence Δ is finitely satisfiable. For every n, α_n or $\neg\alpha_n$ lies in Δ. So if $\alpha \notin \Delta$, then $\neg\alpha \in \Delta$, and $\Delta \cup \{\alpha\}$ contains the unsatisfiable set $\{\alpha, \neg\alpha\}$. Thus Δ is maximal.

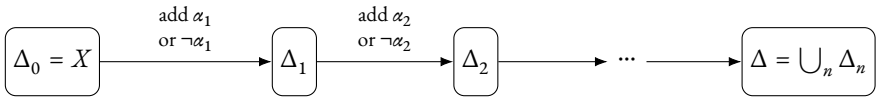


Figure 2.1 The construction of Δ. At step n we add α_n if this keeps the set finitely satisfiable, and $\neg\alpha_n$ otherwise.

Lemma 2.2.5 *Let Δ be maximal finitely satisfiable. For all formulas α and β:*

(a) exactly one of α and $\neg\alpha$ is in Δ;

(b) $\alpha \vee \beta \in \Delta$ if and only if $\alpha \in \Delta$ or $\beta \in \Delta$.

Proof. (a) Not both, since $\{\alpha, \neg\alpha\}$ is not satisfiable. If neither were in Δ, then by maximality neither $\Delta \cup \{\alpha\}$ nor $\Delta \cup \{\neg\alpha\}$ would be finitely satisfiable, contradicting Lemma 2.2.3.

(b) Suppose $\alpha \vee \beta \in \Delta$ but $\alpha, \beta \notin \Delta$. By (a), $\neg\alpha, \neg\beta \in \Delta$, and the finite set $\{\alpha \vee \beta, \neg\alpha, \neg\beta\} \subseteq \Delta$ is not satisfiable. Conversely, suppose $\alpha \in \Delta$ but $\alpha \vee \beta \notin \Delta$. By (a), $\neg(\alpha \vee \beta) \in \Delta$, and $\{\alpha, \neg(\alpha \vee \beta)\} \subseteq \Delta$ is not satisfiable. The case $\beta \in \Delta$ is the same.

The valuation of a maximal set For a maximal finitely satisfiable set Δ, define the valuation v_Δ by $v_\Delta(p) = \top$ if and only if $p \in \Delta$.

Lemma 2.2.6 (Truth lemma) *Let Δ be maximal finitely satisfiable. For every formula α, $v_\Delta \vDash \alpha$ if and only if $\alpha \in \Delta$.*

Proof. By structural induction. For an atomic proposition this is the definition of v_Δ. For $\neg\beta$: $v_\Delta \vDash \neg\beta$ if and only if $v_\Delta \nvDash \beta$, if and only if $\beta \notin \Delta$ by induction, if and only if $\neg\beta \in \Delta$ by Lemma 2.2.5(a). For $\beta \vee \gamma$: $v_\Delta \vDash \beta \vee \gamma$ if and only if $v_\Delta \vDash \beta$ or $v_\Delta \vDash \gamma$, if and only if $\beta \in \Delta$ or $\gamma \in \Delta$, if and only if $\beta \vee \gamma \in \Delta$ by Lemma 2.2.5(b).

***Exercise** 2.2.7* Let Δ be maximal finitely satisfiable. Show that $\alpha \wedge \beta \in \Delta$ if and only if $\alpha \in \Delta$ and $\beta \in \Delta$, and that $\alpha \supset \beta \in \Delta$ if and only if $\alpha \notin \Delta$ or $\beta \in \Delta$.

2.3 Proof of the compactness theorem

Proof of Theorem 2.1.2. Let X be finitely satisfiable. By Lemma 2.2.4, X is contained in a maximal finitely satisfiable set Δ. By Lemma 2.2.6, $v_\Delta \models \Delta$, and so $v_\Delta \models X$.

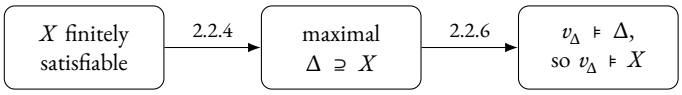


Figure 2.2 The steps in the proof of Theorem 2.1.2. Each arrow is labelled by the lemma it uses.

***Exercise** 2.3.1* Show that every maximal finitely satisfiable set is Δ_v for some valuation v, as in Example 2.2.2.

***Exercise** 2.3.2* Let $\mathcal{P}$ be infinite and $X = \{p_1\} \cup \{p_i \supset p_{i+1} : i \geq 1\}$. Show that exactly one maximal finitely satisfiable set contains X.

***Exercise** 2.3.3* (Kőnig's lemma) A rooted tree is *finitely branching* if every node has finitely many children. Use Theorem 2.1.2 to show that every infinite, finitely branching tree has an infinite path from the root.[4] (Take an atom p_x for each node x, read as "x is on the path".)

***Exercise** 2.3.4* Where does the proof use that $\mathcal{P}$ is countable? Where does it use that formulas are finite strings?

2.4 Compactness and consequence

By definition (Section 1.4), $X \models \alpha$ means that every valuation satisfying X satisfies α. A consequence of an infinite set follows from finitely many of its members.

Lemma 2.4.1 *For all $Z \subseteq \Phi$ and $\beta \in \Phi$, $Z \models \beta$ if and only if $Z \cup \{\neg\beta\}$ is not satisfiable.*

Proof. A valuation v satisfies $Z \cup \{\neg\beta\}$ exactly when it satisfies Z and $v(\beta) = \bot$. Such a valuation exists if and only if $Z \not\models \beta$.

For finite nonempty Z this is Exercise 1.4.1.

Theorem 2.4.2 *Let $X \subseteq \Phi$ and $\alpha \in \Phi$. Then $X \models \alpha$ if and only if $Y \models \alpha$ for some $Y \subseteq_{\text{fin}} X$.*

Proof. ($\Leftarrow$) Let $Y \subseteq_{\text{fin}} X$ with $Y \models \alpha$. If $v \models X$, then $v \models Y$, so $v \models \alpha$.

($\Rightarrow$) Suppose $X \models \alpha$. By Lemma 2.4.1, $X \cup \{\neg\alpha\}$ is not satisfiable. By Theorem 2.1.2, some $Z \subseteq_{\text{fin}} X \cup \{\neg\alpha\}$ is not satisfiable. Let $Y = Z \setminus \{\neg\alpha\}$, a finite subset of X. Then $Y \cup \{\neg\alpha\} \supseteq Z$ is not satisfiable either, so $Y \models \alpha$ by Lemma 2.4.1.

[4] D. Kőnig, "Über eine Schlussweise aus dem Endlichen ins Unendliche", *Acta Scientiarum Mathematicarum (Szeged)* 3 (1927), 121–130; the journal then had a longer Latin title. The title translates as "On a method of inference from the finite to the infinite", which describes compactness as well.

2.5 A topological view

This section assumes topology and is beyond the course. It explains the name of the theorem. Marshall Stone, who connected Boolean algebras, and so propositional logic, with compact spaces, wrote:[5]

> *A cardinal principle of modern mathematical research may be stated as a maxim: "One must always topologize."*

Give $\{\top, \bot\}$ the discrete topology, and give the set $\{\top, \bot\}^{\mathscr{P}}$ of all valuations the product topology. By Tychonoff's theorem this space is compact.[6] Each formula α determines the set $[\![\alpha]\!] = \{v : v \vDash \alpha\}$, which is closed (and open), since $v(\alpha)$ depends only on the finitely many values $v(p)$ for $p \in \mathit{Voc}(\alpha)$. A set X is satisfiable exactly when $\bigcap_{\alpha \in X} [\![\alpha]\!]$ is nonempty, and finitely satisfiable exactly when the family $\{[\![\alpha]\!] : \alpha \in X\}$ has the finite intersection property. In a compact space every family of closed sets with the finite intersection property has nonempty intersection, which is Theorem 2.1.2. This argument works for a set of atomic propositions of any size, with Tychonoff's theorem resting on the axiom of choice.

[5] M. H. Stone, "The representation of Boolean algebras", *Bulletin of the American Mathematical Society* 44 (1938), 807–816, an expository address. Stone had shown in 1936 that every Boolean algebra is isomorphic to the algebra of clopen subsets of a compact Hausdorff space, now called its Stone space; the space of valuations used here is a special case.

[6] A. Tychonoff, "Über die topologische Erweiterung von Räumen", *Mathematische Annalen* 102 (1930), 544–561. For products of finite spaces, as here, the theorem can also be proved with Kőnig's lemma when $\mathscr{P}$ is countable.

Applications I

Good mathematicians see analogies between theorems or theories; the very best ones see analogies between analogies.
Stefan Banach, as recalled by Stanisław Ulam

In each application, an infinite set X of formulas describes an infinite object, and a valuation satisfying X gives the object we want. A finite subset of X mentions only a finite part of the object, where the finite version of the result applies. So X is finitely satisfiable, and by compactness it is satisfiable.

Each proof uses its own atomic propositions, named to suit the problem, such as r_u for "vertex u is red". When there are countably many, we can list them as $p_1, p_2, \dots$, which puts us in the setting of Chapter 2. Each proof therefore checks that its atoms are countable.

3.1 Graph colouring

A *graph* $G = (V, E)$ consists of a set V of *vertices* and a set E of *edges*, each a set $\{u, w\}$ of two distinct vertices. For $k \geq 1$, a *k-colouring* of G is a function $c : V \to \{1, \dots, k\}$ with $c(u) \neq c(w)$ for every edge $\{u, w\}$. For $V_0 \subseteq V$, the *subgraph induced by* V_0 has vertex set V_0 and the edges of G with both ends in V_0. A finite graph is *planar* if it can be drawn in the plane without edges crossing.

Theorem 3.1.1 (Four colour theorem; Appel and Haken, 1976) *Every finite planar graph has a 4-colouring.*

Francis Guthrie posed the four colour problem in 1852, and Kenneth Appel and Wolfgang Haken proved the theorem in 1976.[1] We omit the proof, and ask whether the result holds for an infinite graph, such as the one in Figure 3.1. Colouring such a graph piece by piece may fail, since a colouring chosen for one finite piece may not extend to a larger piece. In the proof below, no colouring of a finite piece is fixed in advance.

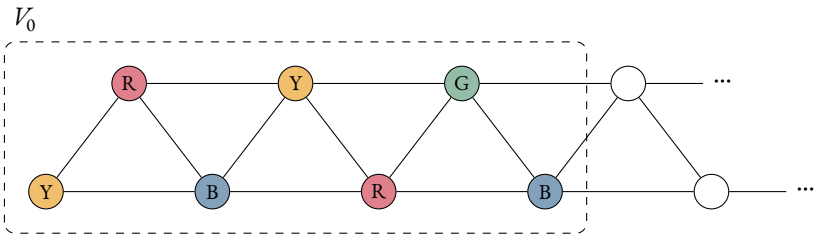


Figure 3.1 Part of an infinite planar graph, a strip of triangles that continues to the right. The finite set V_0 induces a planar subgraph, which has a 4-colouring (R, Y, B, G).

Theorem 3.1.2 *Let $G = (V, E)$ be a graph with V countable, and suppose every finite induced subgraph of G is planar. Then G has a 4-colouring.*

Proof. Take atomic propositions r_u, y_u, b_u, g_u for each $u \in V$, read as "u is red", "yellow", "blue", "green"; there are countably many of them. Let X consist of these formulas.

(a) For each $u \in V$: $r_u \vee y_u \vee b_u \vee g_u$.

(b) For each $u \in V$ and each pair of distinct colours: the formulas $\neg(r_u \wedge y_u)$, $\neg(r_u \wedge b_u)$, ..., $\neg(b_u \wedge g_u)$, one for each of the six pairs.

(c) For each edge $\{u, w\}$: $\neg(r_u \wedge r_w)$, $\neg(y_u \wedge y_w)$, $\neg(b_u \wedge b_w)$, $\neg(g_u \wedge g_w)$.

A valuation satisfying X makes exactly one colour atom true at each vertex by (a) and (b), and so defines a function $c : V \to \{\text{red, yellow, blue, green}\}$; by (c), c is a 4-colouring.

By Theorem 2.1.2 it is enough to show that every $Y \subseteq_{\text{fin}} X$ is satisfiable. Let V_0 be the finite set of vertices whose atoms occur in Y. The subgraph induced by V_0 is planar, so it has a 4-colouring c_0. Let v be the valuation that makes the atom for colour $c_0(u)$ true at each $u \in V_0$ and every other atom false. Each formula of Y mentions only vertices in V_0, and each formula of type (c) in Y comes from an edge of the induced subgraph, so $v \models Y$.

***Exercise** 3.1.3* Write out the formulas of types (a), (b) and (c) for the graph with two vertices u, w and one edge $\{u, w\}$, and give a valuation that satisfies them all.

[1] Guthrie's question reached Augustus De Morgan, who described it in a letter to William Rowan Hamilton dated 23 October 1852. The proof is K. Appel and W. Haken, "Every planar map is four colorable", *Bulletin of the American Mathematical Society* 82 (1976), 711–712. It reduces the problem to a large number of configurations checked by computer. N. Robertson, D. Sanders, P. Seymour and R. Thomas gave a simpler proof in 1997, still using a computer, and G. Gonthier checked a proof in the Coq proof assistant in 2005.

***Exercise** 3.1.4* Show that the proof fails without the formulas of type (a): find a valuation that satisfies all formulas of types (b) and (c) for every graph. Show that the formulas of type (b) can be dropped. (Hint: from a valuation satisfying (a) and (c), choose one true colour atom at each vertex.)

The de Bruijn–Erdős theorem Nothing in the proof used planarity except to colour each finite piece. The same argument, with atoms $p_{u,i}$ for $u \in V$ and $1 \le i \le k$, read as "u has colour i", proves the following; there are countably many such atoms, since $V \times \{1, \dots, k\}$ is countable.

Theorem 3.1.5 (de Bruijn and Erdős, 1951) *Let G be a graph with countably many vertices, and let $k \ge 1$. If every finite induced subgraph of G has a k-colouring, then so does G.*

Theorem 3.1.2 is the case $k = 4$, with the four colour theorem supplying the colourings of the finite subgraphs. Nicolaas de Bruijn and Paul Erdős published the theorem in 1951.[2] The de Bruijn–Erdős theorem also holds for graphs with uncountably many vertices, by the topological form of compactness in Section 2.5.

3.2 Hall's marriage theorem

Let B be a set of boys and $\mathcal{G}$ a set of girls, and suppose each boy b knows a set $N(b) \subseteq \mathcal{G}$ of girls. For $S \subseteq B$ let $N(S) = \bigcup_{b \in S} N(b)$. A *matching* is an injective function $f : B \to \mathcal{G}$ with $f(b) \in N(b)$ for every b: each boy marries a girl he knows, and no girl marries twice (Figure 3.2(a)). If a matching exists, every finite set S of boys knows at least $|S|$ girls. This is *Hall's condition*:

$$|N(S)| \ge |S| \quad \text{for every finite } S \subseteq B.$$

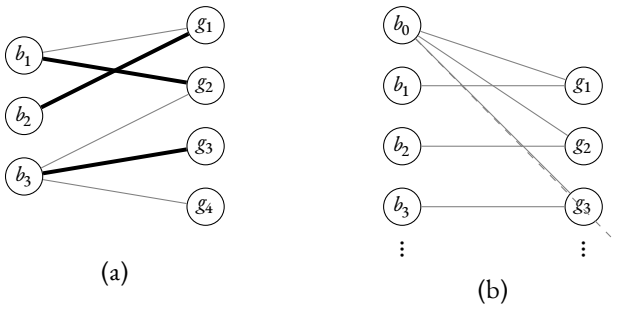


Figure 3.2 (a) Thin lines join each boy to the girls he knows; thick lines form a matching. (b) The boy b_0 knows every girl, and b_i knows only g_i. Hall's condition holds, but there is no matching (see Exercise 3.2.4).

[2] N. G. de Bruijn and P. Erdős, "A colour problem for infinite graphs and a problem in the theory of relations", *Indagationes Mathematicae* 13 (1951), 371–373.

***Exercise** 3.2.1* Check Hall's condition for the boys in Figure 3.2(a), and find a matching different from the one drawn.

Theorem 3.2.2 (P. Hall, 1935) *If B is finite, each $N(b)$ is finite, and Hall's condition holds, then a matching exists.*

Philip Hall proved this in 1935.[3] We omit its proof; Marshall Hall Jr. extended it to infinitely many boys in 1948.[4]

Theorem 3.2.3 (M. Hall Jr., 1948) *Let B be countable, let each $N(b)$ be finite, and suppose Hall's condition holds. Then a matching exists.*

Proof. Take an atomic proposition $m_{b,g}$ for each $b \in B$ and $g \in N(b)$, read as "b marries g". There are countably many, since B is countable and each $N(b)$ is finite. Let X consist of the following formulas.

(a) For each b: the disjunction of $m_{b,g}$ over $g \in N(b)$. This is a finite disjunction, and $N(b)$ is nonempty by Hall's condition for $S = \{b\}$.

(b) For each b and distinct $g, g' \in N(b)$: $\neg(m_{b,g} \wedge m_{b,g'})$.

(c) For each g and distinct b, b' with $g \in N(b) \cap N(b')$: $\neg(m_{b,g} \wedge m_{b',g})$.

A valuation satisfying X defines $f(b)$ as the unique g with $m_{b,g}$ true, by (a) and (b), and f is injective by (c).

Let $Y \subseteq_{\text{fin}} X$, and let B_0 be the finite set of boys whose atoms occur in Y. Hall's condition holds for the boys in B_0, so the finite theorem gives a matching f_0 of B_0. The valuation that makes $m_{b,g}$ true exactly when $b \in B_0$ and $f_0(b) = g$ satisfies Y. By Theorem 2.1.2, X is satisfiable.

***Exercise** 3.2.4* Let $B = \{b_0, b_1, b_2, \dots\}$ and $\mathcal{G} = \{g_1, g_2, \dots\}$, where b_0 knows every girl and b_i knows only g_i for $i \geq 1$, as in Figure 3.2(b). Show that Hall's condition holds and no matching exists. Which step of the proof fails?

3.3 Tiling the plane

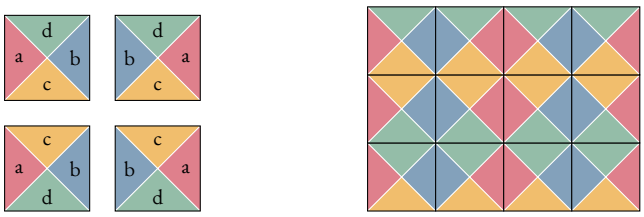
A *Wang tile* is a unit square with a colour on each of its four edges. Fix a finite set W of tiles, with copies of each tile available without limit and without rotation. For $w \in W$, write $\mathit{left}(w), \mathit{right}(w), \mathit{top}(w)$ and $\mathit{bottom}(w)$ for its edge colours. A *tiling* of a set $R \subseteq \mathbb{Z}^2$ of cells is a function $\tau : R \to W$ such that neighbouring edges match: whenever the cells involved lie in R,

$$\mathit{right}(\tau(x,y)) = \mathit{left}(\tau(x+1,y)), \qquad \mathit{top}(\tau(x,y)) = \mathit{bottom}(\tau(x,y+1)).$$

[3] P. Hall, "On representatives of subsets", *Journal of the London Mathematical Society* 10 (1935), 26–30. Hall stated it in terms of *systems of distinct representatives*: choosing a different element from each of a family of finite sets. It is closely related to Kőnig's theorem on matchings in bipartite graphs.

[4] M. Hall Jr., "Distinct representatives of subsets", *Bulletin of the American Mathematical Society* 54 (1948), 922–926.

We say W *tiles* R if some tiling of R exists. Hao Wang introduced these tiles in 1961 and asked whether one can decide, given W, whether W tiles the plane.[5] Robert Berger showed in 1966 that no algorithm decides this.[6]



the tile set W part of a tiling of $\mathbb{Z}^2$

Figure 3.3 A set W of four tiles with edge colours a, b, c, d, and part of a tiling of the plane by W. Neighbouring edges have the same colour; the pattern repeats in both directions.

***Exercise** 3.3.1* Describe the tiling in Figure 3.3: give $\tau(x, y)$ in terms of whether x and y are even or odd.

Theorem 3.3.2 *If W tiles the square $\{-n, \dots, n\}^2$ for every n, then W tiles $\mathbb{Z}^2$.*

Proof. Take an atomic proposition $t_{x,y,w}$ for each cell $(x, y) \in \mathbb{Z}^2$ and tile $w \in W$, read as "cell (x, y) holds tile w"; there are countably many, since $\mathbb{Z}^2 \times W$ is countable. Let X consist of the following formulas, for all x, y.

(a) The disjunction of $t_{x,y,w}$ over $w \in W$.

(b) $\neg(t_{x,y,w} \wedge t_{x,y,w'})$ for distinct $w, w' \in W$.

(c) $\neg(t_{x,y,w} \wedge t_{x+1,y,w'})$ whenever $\mathit{right}(w) \neq \mathit{left}(w')$.

(d) $\neg(t_{x,y,w} \wedge t_{x,y+1,w'})$ whenever $\mathit{top}(w) \neq \mathit{bottom}(w')$.

A valuation satisfying X defines a tiling of $\mathbb{Z}^2$: let $\tau(x, y)$ be the unique w with $t_{x,y,w}$ true, which exists by (a) and (b); edges match by (c) and (d). A finite $Y \subseteq_{\text{fin}} X$ mentions only cells in some square $\{-n, \dots, n\}^2$, and a tiling of that square gives a valuation satisfying Y. By Theorem 2.1.2, X is satisfiable.

***Exercise** 3.3.3* Show that W tiles $\mathbb{Z}^2$ if and only if it tiles the quadrant $\{0, 1, 2, \dots\}^2$. (Hint: tilings can be translated, and the quadrant contains a translate of every square.)

Further exercises

[5] H. Wang, "Proving theorems by pattern recognition II", *Bell System Technical Journal* 40 (1961), 1–41. Wang showed that such an algorithm would exist if every tile set that tiles the plane also tiles it periodically.

[6] R. Berger, "The undecidability of the domino problem", *Memoirs of the American Mathematical Society* 66 (1966). Berger's proof produced a tile set that tiles the plane but never periodically; it had 20,426 tiles. E. Jeandel and M. Rao found an aperiodic set of 11 Wang tiles in 2015, and showed that no smaller set exists.

***Exercise** 3.3.4* A *partial order* $\leq$ on a set A is reflexive, antisymmetric and transitive; it is *linear* if any two elements are comparable. Every partial order on a finite set extends to a linear order. Use compactness to show that every partial order on a countable set extends to a linear order. (Use atoms $p_{a,b}$ for $a, b \in A$, read as "a comes before b in the new order".)

First-Order Logic

"When I *use a word," Humpty Dumpty said, in rather a scornful tone,*
"it means just what I choose it to mean, neither more nor less."
Lewis Carroll, *Through the Looking-Glass*

In Chapter 3, each infinite object needed its own atomic propositions, one for each vertex or cell. In first-order logic, one sentence with variables and quantifiers can state a property of every graph or every group. A structure such as a graph or a group is a set of elements with relations, functions and constants on it. First-order logic adds symbols for these to the propositional connectives, together with a quantifier $\exists$ ("there exists") over elements. Quantifiers first appear in Gottlob Frege's *Begriffsschrift* of 1879.[1] This chapter follows Chapter 4 of Mukund and Suresh's notes.

An example Take a binary relation symbol E for the edges of a graph. Using notation defined below, the sentence

$$\forall x \, \forall y \, (E(x, y) \supset E(y, x))$$

says that edges are symmetric, and $\forall x \, \exists y \, \exists z \, (E(x, y) \wedge E(x, z) \wedge \neg(y \approx z))$ says that every vertex has at least two neighbours. A graph *satisfies* a sentence if the sentence is true when E is read as its edge relation.

[1] G. Frege, *Begriffsschrift, eine der arithmetischen nachgebildete Formelsprache des reinen Denkens*, Louis Nebert, Halle, 1879. Frege's notation was two-dimensional and is no longer used; the quantifier symbols $\forall$ and $\exists$ came later: $\forall$ from Gentzen (1935) and $\exists$ from Peano (1897).

4.1 Syntax

Definition 4.1.1 (Languages) A *first-order language* is a triple $L = (R, F, C)$ of sets of *relation symbols*, *function symbols* and *constant symbols*, together with an *arity* $n \geq 1$ for each relation and function symbol, the number of arguments it takes. We also fix a countable set Var of *variables*, written x, y, z. The symbols of L are those in R, F, C and Var, the equality symbol $\approx$, the connectives $\neg$ and $\vee$, the quantifier $\exists$, the comma and the two parentheses; all are distinct.

Until Section 5.1 we assume that R, F and C are countable. Binary symbols such as $<$ and $+$ may be written between their arguments.

Definition 4.1.2 (Terms) The *terms* over L form the smallest set of strings containing every constant and every variable, and containing $f(t_1, \dots, t_n)$ whenever $f \in F$ has arity n and $t_1, \dots, t_n$ are terms. A term with no variables is *closed*.

Definition 4.1.3 (Formulas) An *atomic formula* is a string $r(t_1, \dots, t_n)$, with $r \in R$ of arity n and terms $t_1, \dots, t_n$, or $t_1 \approx t_2$, with terms t_1, t_2. The set Φ_L of *formulas* is the smallest set of strings that contains the atomic formulas, contains $(\neg\varphi)$ and $(\varphi \vee \psi)$ whenever it contains φ and ψ, and contains $\exists x\, \varphi$ whenever it contains φ and $x \in Var$.

We write φ, ψ, χ for first-order formulas, keeping α, β, γ for propositional ones, and $\approx$ for equality, since $\equiv$ already means *if and only if*; Mukund and Suresh write $\equiv$. As in Chapter 1, every term and every formula is built in exactly one way, so we may define functions on them by recursion and prove facts about them by structural induction; we drop parentheses by the same conventions. The connectives $\wedge$, $\supset$, $\equiv$ are abbreviations as before, and $\forall x\, \varphi \stackrel{\text{def}}{=} \neg\exists x\, \neg\varphi$.

Definition 4.1.4 (Free variables) For a term t, let $Var(t)$ be the set of variables occurring in t. The set $FV(\varphi)$ of *free variables* of a formula is defined by recursion:

$$FV(r(t_1, \dots, t_n)) = Var(t_1) \cup \dots \cup Var(t_n), \qquad FV(t_1 \approx t_2) = Var(t_1) \cup Var(t_2),$$
$$FV(\neg\varphi) = FV(\varphi), \qquad FV(\varphi \vee \psi) = FV(\varphi) \cup FV(\psi), \qquad FV(\exists x\, \varphi) = FV(\varphi) \setminus \{x\}.$$

A *sentence* is a formula φ with $FV(\varphi) = \emptyset$. We write $\varphi(x)$ for a formula with $FV(\varphi) \subseteq \{x\}$.

For example, $FV(E(x, y) \vee \exists x\, E(x, x)) = \{x, y\}$: the x in $E(x, y)$ is free, and the two in $E(x, x)$ are bound by $\exists x$.

Definition 4.1.5 (Substitution) Let x be a variable and t a closed term. For a term u, the term $u[t/x]$ replaces each occurrence of x in u by t. For a formula χ, the formula $\chi[t/x]$ is defined by recursion:

$$r(t_1, \dots, t_n)[t/x] = r(t_1[t/x], \dots, t_n[t/x]), \qquad (t_1 \approx t_2)[t/x] = (t_1[t/x] \approx t_2[t/x]),$$
$$(\neg\varphi)[t/x] = \neg(\varphi[t/x]), \qquad (\varphi \vee \psi)[t/x] = \varphi[t/x] \vee \psi[t/x],$$
$$(\exists x\, \varphi)[t/x] = \exists x\, \varphi, \qquad (\exists y\, \varphi)[t/x] = \exists y\, (\varphi[t/x]) \text{ for } y \neq x.$$

We write $\varphi(t)$ for $\varphi(x)[t/x]$.

So $\chi[t/x]$ replaces the free occurrences of x in χ by t. Since t is closed, no variable of t can be captured by a quantifier, and by structural induction $FV(\chi[t/x]) = FV(\chi) \setminus \{x\}$; in particular $\varphi(t)$ is a sentence.

***Exercise** 4.1.6* Find the free variables of $\exists x\,(E(x,y) \wedge \exists y\, E(y,z))$. Is it a sentence? Compute $(E(x,y) \vee \exists x\, E(x,x))[c/x]$ for a constant c.

4.2 Semantics

Definition 4.2.1 (Structures) A *structure* for L is a pair $M = (D, \iota)$, where D is a nonempty set, the *domain* of M, and ι assigns a relation $r^M = \iota(r) \subseteq D^n$ to each $r \in R$ of arity n, a function $f^M = \iota(f) : D^n \to D$ to each $f \in F$ of arity n, and an element $c^M = \iota(c) \in D$ to each $c \in C$.

Example 4.2.2 Let L have one binary relation symbol E. The structure M with $D = \{1,2,3\}$ and $E^M = \{(1,2),(2,1),(2,3),(3,2)\}$ is the path in Figure 4.1. It satisfies the symmetry sentence of the opening example. It does not satisfy the second sentence there, since vertex 1 has only one neighbour.



Figure 4.1 The structure of Example 4.2.2: each line stands for the two pairs (s,s') and (s',s) in E^M.

To evaluate a formula with free variables we must also say what the variables stand for.

Definition 4.2.3 (Assignments and values) An *assignment* in M is a function $\sigma : Var \to D$. For $x \in Var$ and $s \in D$, the assignment $\sigma[x \mapsto s]$ sends x to s and agrees with σ elsewhere. The *value* $t^{M,\sigma} \in D$ of a term t is defined by recursion:

$$c^{M,\sigma} = c^M, \qquad x^{M,\sigma} = \sigma(x), \qquad f(t_1,\dots,t_n)^{M,\sigma} = f^M(t_1^{M,\sigma},\dots,t_n^{M,\sigma}).$$

By structural induction, the value of a closed term t does not depend on σ; we write it t^M.

Definition 4.2.4 (Satisfaction) The relation $(M,\sigma) \vDash \varphi$, read "$\varphi$ is true in M under σ", is defined by recursion on φ.

- $(M,\sigma) \vDash t_1 \approx t_2$ if $t_1^{M,\sigma} = t_2^{M,\sigma}$.
- $(M,\sigma) \vDash r(t_1,\dots,t_n)$ if $(t_1^{M,\sigma},\dots,t_n^{M,\sigma}) \in r^M$.
- $(M,\sigma) \vDash \neg\varphi$ if $(M,\sigma) \nvDash \varphi$.
- $(M,\sigma) \vDash \varphi \vee \psi$ if $(M,\sigma) \vDash \varphi$ or $(M,\sigma) \vDash \psi$.
- $(M,\sigma) \vDash \exists x\,\varphi$ if $(M,\sigma[x \mapsto s]) \vDash \varphi$ for some $s \in D$.

***Exercise** 4.2.5* Does the path M of Example 4.2.2 satisfy $\exists x\, \forall y\, (\neg(x \approx y) \supset E(x,y))$? Does it satisfy $\forall x\, \exists y\, E(x,y)$?

Lemma 4.2.6 *Let M be a structure for L.*

(1) If assignments σ and σ' agree on $FV(\varphi)$, then $(M,\sigma) \vDash \varphi$ if and only if $(M,\sigma') \vDash \varphi$.

(2) Substitution lemma. *For every formula χ, variable x, assignment σ and closed term t,*

$$(M,\sigma) \vDash \chi[t/x] \quad \textit{if and only if} \quad (M, \sigma[x \mapsto t^M]) \vDash \chi.$$

(3) If M' is a structure for a larger language that agrees with M on the symbols of L, then M and M' satisfy the same formulas of L under any assignment.

Proof. Each part is proved by structural induction on the formula; see Exercise 4.2.9. Part (1) is the first-order form of Proposition 1.3.3.

By (1), whether a sentence φ is true in M does not depend on the assignment, and we write $M \vDash \varphi$. By (2), $M \vDash \varphi(t)$ if and only if $(M, \sigma[x \mapsto t^M]) \vDash \varphi(x)$ for any assignment σ.

Definition 4.2.7 (Models and consequence) Let X be a set of sentences. A structure M *satisfies* X, written $M \vDash X$, if $M \vDash \varphi$ for every $\varphi \in X$; then M is a *model* of X. The set X is *satisfiable* if it has a model, and, as in Section 2.1, *finitely satisfiable* if every $Y \subseteq_{\text{fin}} X$ is satisfiable. For a sentence φ, we write $X \vDash \varphi$ if every model of X satisfies φ.

For $n \geq 1$, let

$$\lambda_n = \exists x_1 \cdots \exists x_n \bigwedge_{i<j} \neg(x_i \approx x_j)$$

(for $n = 1$, read the empty conjunction as $x_1 \approx x_1$), which holds in M if and only if D has at least n elements.

***Exercise** 4.2.8* Show that $(M,\sigma) \vDash \forall x\, \varphi$ if and only if $(M, \sigma[x \mapsto s]) \vDash \varphi$ for every $s \in D$. Check that λ_n holds in M if and only if D has at least n elements.

***Exercise** 4.2.9* Prove Lemma 4.2.6 by structural induction, for all assignments at once. (In part (2), for $\exists y\, \chi$, treat $y = x$ and $y \neq x$ separately.)

Compactness The compactness theorem of Chapter 2 has a first-order version. In 1928 David Hilbert and Wilhelm Ackermann asked whether every first-order sentence that is true in all structures can be proved.[2] Gödel's doctoral thesis answered yes, and compactness for countable languages followed from it. Mal'cev extended compactness to languages of any size, as described in Section 2.1.

[2] D. Hilbert and W. Ackermann, *Grundzüge der theoretischen Logik*, Springer, Berlin, 1928.

Theorem 4.2.10 (Compactness; Gödel, 1930, and Mal'cev, 1936) *A set X of sentences of L is satisfiable if and only if it is finitely satisfiable.*

If $M \models X$ and $Y \subseteq X$, then $M \models Y$. For the converse, as in Chapter 2, we extend a finitely satisfiable set as far as possible and read a model off the result; Section 4.3 gives the construction and Section 4.4 the proof.

4.3 Henkin's construction

This section parallels Section 2.2. The construction is due to Leon Henkin, who used it in 1949 to give a new proof of Gödel's completeness theorem.[3] Here it proves compactness directly. The model will be made of closed terms, so the language first needs enough constants. Suppose L has one constant c and one unary relation symbol r. The set $\{\exists x\, r(x),\ \neg r(c)\}$ is satisfiable, but the only closed term is c, so a model made of closed terms has a single element c, which is not in r. A new constant naming an element of r removes the problem.

The witnessing expansion Let $L_0 = L$. Given L_n, form L_{n+1} by adding a new constant c_φ for each formula $\varphi(x)$ of L_n that does not yet have one. Here $\varphi(x)$ is a formula paired with a variable x that is its only free variable, if any; the notation c_φ leaves x out, and a sentence gets one constant for each variable. A formula that mentions a new constant needs a constant of its own, so the step is repeated. Let L_H be the union of the L_n. Every formula of L_H uses finitely many symbols, so it lies in some L_n, and every formula $\varphi(x)$ of L_H has its constant c_φ. Each L_n has countably many symbols, so countably many formulas, and hence L_H has countably many symbols. The *Henkin axioms* are the sentences

$$\exists x\, \varphi(x) \supset \varphi(c_\varphi)$$

for all formulas $\varphi(x)$ of L_H; let H be the set of them.

Lemma 4.3.1 *If X is a finitely satisfiable set of sentences of L, then $X \cup H$ is finitely satisfiable.*

Proof. Let Z be a finite subset of $X \cup H$. The finite set $Z \cap X$ has a model M. Extend M to $L_1, L_2, \ldots$ in turn: when c_φ is added to L_{n+1}, let its value be some s with $(M_n, \sigma[x \mapsto s]) \models \varphi(x)$ for an assignment σ, if there is one, and any element otherwise (by Lemma 4.2.6(1) the choice of σ does not matter), where M_n is the structure already built for L_n. Let M_H be the structure for L_H that agrees with M_n on the symbols of L_n for every n. By Lemma 4.2.6(3), M_H satisfies $Z \cap X$. For a Henkin axiom whose constant c_φ was added to L_{n+1}, Lemma 4.2.6(3) lets us evaluate $\exists x\, \varphi(x)$ in M_n, and the substitution lemma gives $M_H \models \varphi(c_\varphi)$ whenever $M_H \models \exists x\, \varphi(x)$. So $M_H \models Z$.

[3] L. Henkin, "The completeness of the first-order functional calculus", *Journal of Symbolic Logic* 14 (1949), 159–166. The paper came from Henkin's 1947 doctoral thesis at Princeton, written under Alonzo Church. Henkin's method is now the standard proof of completeness, and is used in many other logics.

Maximal sets As in Section 2.2, a finitely satisfiable set Σ of sentences of L_H is *maximal* if, for every sentence $\varphi \notin \Sigma$, the set $\Sigma \cup \{\varphi\}$ is not finitely satisfiable.

Example 4.3.2 As in Example 2.2.2, for a structure M for L_H the set $\Sigma_M = \{\varphi : M \models \varphi\}$ of sentences of L_H is maximal finitely satisfiable. If M interprets the constants c_φ as in the proof of Lemma 4.3.1, then $H \subseteq \Sigma_M$. The proof of compactness goes the other way, and builds a structure from a maximal finitely satisfiable set.

Lemma 4.3.3 (Lindenbaum) *Every finitely satisfiable set of sentences of L_H is contained in a maximal finitely satisfiable set.*

Proof. The proofs of Lemmas 2.2.3 and 2.2.4 work with structures in place of valuations. They need a list $\varphi_1, \varphi_2, \ldots$ of all sentences of L_H, which exists because L_H has countably many symbols.

Lemma 4.3.4 *Let Σ be a maximal finitely satisfiable set of sentences of L_H with $H \subseteq \Sigma$. For all sentences φ, ψ of L_H:*

(a) exactly one of φ and $\neg\varphi$ is in Σ;

(b) $\varphi \vee \psi \in \Sigma$ if and only if $\varphi \in \Sigma$ or $\psi \in \Sigma$;

(c) if $Y \models \psi$ for some finite $Y \subseteq \Sigma$, then $\psi \in \Sigma$;

(d) $\exists x\, \varphi(x) \in \Sigma$ if and only if $\varphi(t) \in \Sigma$ for some closed term t.

Proof. (a) and (b) are proved as in Lemma 2.2.5. (c) Otherwise $\neg\psi \in \Sigma$ by (a), and the finite set $Y \cup \{\neg\psi\} \subseteq \Sigma$ is not satisfiable. (d) If $\exists x\, \varphi(x) \in \Sigma$, then Σ also contains the Henkin axiom $\exists x\, \varphi(x) \supset \varphi(c_\varphi)$, and these two sentences entail $\varphi(c_\varphi)$, so $\varphi(c_\varphi) \in \Sigma$ by (c). Conversely, by the substitution lemma $\{\varphi(t)\} \models \exists x\, \varphi(x)$, so $\varphi(t) \in \Sigma$ gives $\exists x\, \varphi(x) \in \Sigma$ by (c).

The term model Let Σ be as in Lemma 4.3.4. For closed terms t, u of L_H, let $t \sim u$ if $t \approx u \in \Sigma$. By Lemma 4.3.4(c), $\sim$ is an equivalence relation: for instance, $\{t \approx u,\ u \approx w\} \models t \approx w$, so $t \sim u$ and $u \sim w$ give $t \sim w$. Write $[t]$ for the class of t, and define a structure M for L_H:

- D is the set of classes $[t]$ of closed terms of L_H. It is nonempty, since L_H contains the constant $c_{x \approx x}$.
- $c^M = [c]$ for each constant c, and $f^M([t_1], \ldots, [t_n]) = [f(t_1, \ldots, t_n)]$.
- $([t_1], \ldots, [t_n]) \in r^M$ if and only if $r(t_1, \ldots, t_n) \in \Sigma$.

By Lemma 4.3.4(c) again, f^M and r^M do not depend on the chosen representatives, and by induction on t, $t^M = [t]$ for every closed term t.

Lemma 4.3.5 (Truth lemma) *For every sentence φ of L_H, $M \models \varphi$ if and only if $\varphi \in \Sigma$.*

Proof. By induction on the number of occurrences of $\neg$, $\vee$ and $\exists$ in φ. For atomic sentences this is the definition of M: for instance, $M \models t \approx u$ if and only if $[t] = [u]$, if

and only if $t \approx u \in \Sigma$. The cases $\neg$ and $\vee$ follow from the induction hypothesis and Lemma 4.3.4(a) and (b). Let $\varphi = \exists x\, \psi(x)$. Every element of D is t^M for a closed term t, so by the substitution lemma, $M \vDash \varphi$ if and only if $M \vDash \psi(t)$ for some closed term t. The sentence $\psi(t)$ has fewer occurrences than φ, so this holds if and only if $\psi(t) \in \Sigma$ for some t, which by Lemma 4.3.4(d) holds if and only if $\varphi \in \Sigma$.

***Exercise** 4.3.6* Check that $\sim$ is an equivalence relation, that f^M and r^M are well defined, and that $t^M = [t]$ for closed terms t.

***Exercise** 4.3.7* Let L have one unary relation symbol r and no constants. Name two constants of L_1, and one constant of L_2 that is not in L_1.

4.4 Proof of the compactness theorem

Proof of Theorem 4.2.10. The proof parallels the proof of Theorem 2.1.2 in Section 2.3, with Lemmas 4.3.3 and 4.3.5 in place of Lemmas 2.2.4 and 2.2.6; the only new step is adding the Henkin axioms. Let X be finitely satisfiable. By Lemma 4.3.1, $X \cup H$ is finitely satisfiable. By Lemma 4.3.3, $X \cup H$ is contained in a maximal finitely satisfiable set Σ. By Lemma 4.3.5, the term model M of Σ satisfies Σ, and so X. Forgetting the new constants gives a structure for L that satisfies X, by Lemma 4.2.6(3).

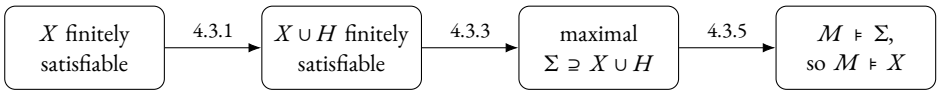


Figure 4.2 The steps in the proof of Theorem 4.2.10, labelled by lemma as in Figure 2.2; M is the term model of Σ.

As in Section 2.4, compactness has a consequence form.

Lemma 4.4.1 *For every set Z of sentences and every sentence φ, $Z \vDash \varphi$ if and only if $Z \cup \{\neg\varphi\}$ is not satisfiable.*

Proof. A structure M satisfies $Z \cup \{\neg\varphi\}$ exactly when it satisfies Z and $M \nvDash \varphi$. Such a structure exists if and only if $Z \nvDash \varphi$.

Theorem 4.4.2 *Let X be a set of sentences and φ a sentence. Then $X \vDash \varphi$ if and only if $Y \vDash \varphi$ for some $Y \subseteq_{\text{fin}} X$.*

Proof. ($\Leftarrow$) Let $Y \subseteq_{\text{fin}} X$ with $Y \vDash \varphi$. If $M \vDash X$, then $M \vDash Y$, so $M \vDash \varphi$.

($\Rightarrow$) Suppose $X \vDash \varphi$. By Lemma 4.4.1, $X \cup \{\neg\varphi\}$ is not satisfiable. By Theorem 4.2.10, some $Z \subseteq_{\text{fin}} X \cup \{\neg\varphi\}$ is not satisfiable. Let $Y = Z \setminus \{\neg\varphi\}$, a finite subset of X. Then $Y \cup \{\neg\varphi\} \supseteq Z$ is not satisfiable either, so $Y \vDash \varphi$ by Lemma 4.4.1.

The term model in the proof of Theorem 4.2.10 has countably many elements, since L_H has countably many closed terms. So a satisfiable set of sentences of a countable language has a countable model. This is the (downward) *Löwenheim–Skolem theorem*,

proved by Leopold Löwenheim in 1915 and Thoralf Skolem in 1920.[4] Applied to set theory, it gives a countable model of axioms that prove the existence of uncountable sets. This is *Skolem's paradox*; it is not a contradiction, because a set that is uncountable inside the model can be countable from outside.

***Exercise** 4.4.3* Let X be a set of sentences with models of every finite size $n \geq 1$. Show that X has an infinite model. (Consider $X \cup \{\lambda_n : n \geq 1\}$.) Conclude that no set of sentences is satisfied by exactly the finite structures.

[4] L. Löwenheim, "Über Möglichkeiten im Relativkalkül", *Mathematische Annalen* 76 (1915), 447–470; T. Skolem, "Logisch-kombinatorische Untersuchungen über die Erfüllbarkeit oder Beweisbarkeit mathematischer Sätze nebst einem Theoreme über dichte Mengen", *Videnskapsselskapets Skrifter, I. Matematisk-naturvidenskabelig Klasse* 4 (1920).

Applications II

Complete disorder is impossible.
Theodore Motzkin

The applications of first-order compactness in this chapter use two arguments. To get a large model, add new constants that must be distinct. To get an infinite model that violates a statement about finite structures, add the sentences λ_n of Section 4.2. We also use the consequence form of compactness, Theorem 4.4.2.

5.1 Large models

Section 4.4 gives countable models, and to get uncountable models from compactness we add uncountably many new constants.

In the proof of Theorem 4.2.10, countability is used only in Lemma 4.3.3, to list the sentences of L_H. For a language of any size, Zorn's lemma gives a maximal finitely satisfiable set instead. The union of a nonempty chain of finitely satisfiable sets containing $X \cup H$ is again such a set, since each of its finite subsets lies in one member of the chain, and $X \cup H$ bounds the empty chain; so Zorn's lemma gives a maximal one. Zorn's lemma is equivalent to the axiom of choice and is beyond the course.[1] With this change, the proof of Theorem 4.2.10 gives first-order compactness for languages of any size.

[1] M. Zorn, "A remark on method in transfinite algebra", *Bulletin of the American Mathematical Society* 41 (1935), 667–670. Kuratowski had proved an equivalent principle in 1922.

Theorem 5.1.1 (Upward Löwenheim–Skolem; Tarski, 1928) *Let X be a set of sentences with an infinite model, and let I be any set. Then X has a model with an injective map from I into its set of elements.*

Proof. Add a new constant c_i for each $i \in I$, and let $X^+ = X \cup \{\neg(c_i \approx c_j) : i, j \in I,\ i \neq j\}$. A finite subset of X^+ mentions finitely many new constants. Interpret them as distinct elements of an infinite model of X, and the other c_i arbitrarily; this satisfies the subset. By compactness, X^+ has a model M. Forgetting the constants c_i gives a model of X with the same elements, and $i \mapsto c_i^M$ is injective.

Taking $I = \mathbb{R}$ shows that every set of sentences with an infinite model has an uncountable model. So no set of first-order sentences with an infinite model has all its models of one size. The upward theorem is due to Alfred Tarski.[2]

***Exercise** 5.1.2* Let Z be the set of sentences true in the countably infinite graph with no edges. Show that every model of Z is an infinite graph with no edges, and that Z has a model with uncountably many elements.

5.2 Ramsey's theorem

Let $G = (V, E)$ be a graph. A set $H \subseteq V$ is a *clique* if any two distinct vertices of H are joined by an edge, and *independent* if no two are. It is *homogeneous* if it is a clique or independent. Ramsey's theorem gives large homogeneous sets in large graphs, however the edges are chosen. Frank Ramsey proved it in 1930, as a step in a paper on the decision problem for a fragment of first-order logic; he died that year, aged 26.[3] Paul Erdős and George Szekeres rediscovered it in 1935 and made it widely known.[4]

Theorem 5.2.1 (Ramsey, 1930) *Every graph with infinitely many vertices has an infinite homogeneous set.*

Proof. Let $V_0 = V$ and choose $u_0 \in V_0$. Suppose V_i is infinite and $u_i \in V_i$. The infinite set $V_i \setminus \{u_i\}$ splits into the neighbours of u_i and the non-neighbours, and one of the two parts is infinite. Let V_{i+1} be an infinite part, colour u_i *red* if V_{i+1} consists of neighbours of u_i and *blue* otherwise, and choose $u_{i+1} \in V_{i+1}$.

For $i < j$ we have $u_j \in V_j \subseteq V_{i+1}$, so u_i is joined to u_j exactly when u_i is red. Infinitely many u_i have the same colour. If infinitely many are red, they form a clique: for red u_i

[2] Tarski proved it in a seminar in 1928. The first printed proof is in Mal'cev's 1936 paper, cited in Section 2.1, which credits Tarski. A. Tarski and R. L. Vaught, "Arithmetical extensions of relational systems", *Compositio Mathematica* 13 (1957), 81–102, gave the version with elementary extensions. The theorem is sometimes called the Löwenheim–Skolem–Tarski theorem. Together with the downward theorem, it shows that a set of sentences of a countable language with an infinite model has models of every infinite size.

[3] F. P. Ramsey, "On a problem of formal logic", *Proceedings of the London Mathematical Society* (2) 30 (1930), 264–286. Ramsey also made lasting contributions to economics and the philosophy of probability.

[4] P. Erdős and G. Szekeres, "A combinatorial problem in geometry", *Compositio Mathematica* 2 (1935), 463–470. They used it to show that among enough points in the plane, no three on a line, some k form a convex polygon. Esther Klein had shown that any five such points include four in convex position, and asked for the general statement; she later married Szekeres, and the problem is known as the *happy ending problem*.

and u_j with $i < j$, u_i is joined to u_j because u_i is red. If infinitely many are blue, they form an independent set in the same way.

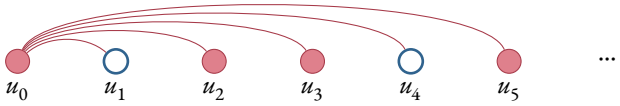


Figure 5.1 The sequence in the proof of Theorem 5.2.1. A red vertex (filled) is joined to every later vertex; a blue vertex (open) to none. Only the edges at u_0 are drawn.

***Exercise** 5.2.2* Let $V = \{0, 1, 2, \dots\}$, with m and n joined when $m + n$ is even. Run the proof of Theorem 5.2.1 with $u_0 = 0$, always choosing the neighbours when they form an infinite set and taking the least element of V_{i+1} as u_{i+1}. What homogeneous set results?

The finite version The finite version states that every sufficiently large graph has a homogeneous set of a given size. It follows from the infinite version by compactness. Let L have one binary relation symbol E, and let Γ consist of the two sentences

$$\forall x\, \neg E(x, x) \quad \text{and} \quad \forall x\, \forall y\, (E(x, y) \supset E(y, x)),$$

whose models are the graphs.

Theorem 5.2.3 (Ramsey, finite version) *For every $k \geq 1$ there is an n such that every graph with n vertices has a homogeneous set with k elements.*

Proof. Let η_k say that there is no homogeneous set with k elements:

$$\neg \exists x_1 \cdots \exists x_k \left(\theta \wedge \bigwedge_{i<j} E(x_i, x_j) \right)$$
$$\wedge\, \neg \exists x_1 \cdots \exists x_k \left(\theta \wedge \bigwedge_{i<j} \neg E(x_i, x_j) \right),$$

where θ is $\bigwedge_{i<j} \neg(x_i \approx x_j)$ and the big conjunctions are finite.

Suppose the theorem fails for some k, necessarily $k \geq 2$: for every n there is a graph with n vertices and no homogeneous set with k elements. Let $X = \Gamma \cup \{\eta_k\} \cup \{\lambda_n : n \geq 1\}$. A finite subset of X contains λ_n only for $n \leq N$, for some N, and a graph with N vertices and no homogeneous set with k elements satisfies it. By Theorem 4.2.10, X has a model. This model is a graph with infinitely many vertices and no homogeneous set with k elements, so no infinite homogeneous set, which contradicts Theorem 5.2.1.

The proof gives no bound on n. The least n that works for k is the *Ramsey number* $R(k)$; exact values are known only for $k \leq 4$, with $R(4) = 18$.[5]

[5] For $R(5)$ only bounds are known: $43 \leq R(5) \leq 46$, the upper bound proved by computer in 2024.

***Exercise** 5.2.4* Show that $R(3) = 6$: every graph with 6 vertices has a homogeneous set with 3 elements, and the 5-cycle (vertices $1, \dots, 5$, with i joined to $i + 1$ and 5 joined to 1) does not. Deduce that no $n \leq 5$ works, by deleting vertices from the 5-cycle.

***Exercise** 5.2.5* Colour each pair of distinct elements of a set with one of m colours. State and prove an infinite version of Ramsey's theorem for m colours, and deduce the finite version by compactness. (Use one relation symbol for each colour, with sentences saying that each pair of distinct elements has exactly one colour.)

5.3 *Fields of large characteristic*

This section assumes familiarity with fields. Let L have constants 0, 1 and binary function symbols $+$, $\cdot$, written between their arguments. The field axioms are finitely many sentences of L; let Σ_F be their set. A field has *characteristic* p, for a prime p, if $1 + \dots + 1 = 0$ with p ones, and characteristic 0 if no such sum is 0; at most one prime has this property, and every field has characteristic 0 or a prime. Let χ_p be the sentence $1 + \dots + 1 \approx 0$ with p ones. The fields of characteristic 0 are the models of $\Sigma_F \cup \{\neg\chi_p : p \text{ prime}\}$.

Theorem 5.3.1 *If a sentence φ of L holds in every field of characteristic* 0*, then there is N such that φ holds in every field of characteristic $p > N$.*

Proof. By assumption $\Sigma_F \cup \{\neg\chi_p : p \text{ prime}\} \models \varphi$. By Theorem 4.4.2, some finite subset Y of this set has $Y \models \varphi$; it mentions $\neg\chi_p$ only for primes $p \leq N$, for some N. A field of characteristic $p > N$ satisfies Σ_F and $\neg\chi_q$ for every prime $q \leq N$, so it satisfies Y, and hence φ.

For example, $\neg(1 + 1 \approx 0)$ holds in every field of characteristic 0 and fails only in characteristic 2.

***Exercise** 5.3.2* Find the least N that Theorem 5.3.1 allows for the sentence $\neg(1+1+1 \approx 0)$.

***Exercise** 5.3.3* Show that no set of sentences of L has as its models exactly the fields of positive characteristic. (Apply compactness to $X \cup \{\neg\chi_p : p \text{ prime}\}$.)

***Exercise** 5.3.4* This exercise assumes familiarity with groups. Let L have a constant e and a binary function symbol $\cdot$. A group is *torsion* if every element has finite order. Show that no set of sentences of L has as its models exactly the torsion groups. (Add a constant c and, for each $n \geq 1$, the sentence $\neg(c^n \approx e)$, where c^n is the product of n copies of c.)

Limits of First-Order Logic

The limits of my language mean the limits of my world.
Ludwig Wittgenstein, *Tractatus Logico-Philosophicus* 5.6

If a set of sentences expressed a property, compactness could give a model of the sentences that lacks the property. This applies to connectivity of graphs and to the natural numbers.

6.1 Connectivity is not first-order

Let L have one binary relation symbol E, and let Γ be the two sentences of Section 5.2 whose models are the graphs. A *walk of length n* from s to s' in a graph is a sequence $s = s_0, s_1, \ldots, s_n = s'$ in which consecutive vertices are joined. A graph is *connected* if any two vertices are joined by a walk.

Theorem 6.1.1 *There is no set X of sentences of L such that a graph satisfies X if and only if it is connected.*

Proof. Suppose X were such a set. Add two constants a and b to L, and for each $n \geq 0$ let δ_n say that there is no walk of length n from a to b:

$$\delta_n = \neg\exists x_0 \cdots \exists x_n \left(x_0 \approx a \wedge x_n \approx b \wedge E(x_0, x_1) \wedge \ldots \wedge E(x_{n-1}, x_n)\right).$$

Let $X^+ = \Gamma \cup X \cup \{\delta_n : n \geq 0\}$.

A finite subset of X^+ mentions δ_n only for $n \leq N$, for some N. The path with vertices $0, 1, \ldots, N+1$, edges between consecutive numbers, $a = 0$ and $b = N+1$, satisfies it: it is a connected graph, so it satisfies Γ and X, and every walk from a to b has length at least $N+1$ (Figure 6.1). By Theorem 4.2.10, X^+ has a model M. Forgetting a and b, M is a graph that satisfies X, so it is connected. But M satisfies every δ_n, so there is no walk from a^M to b^M, a contradiction.

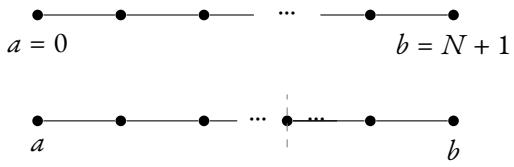


Figure 6.1 Top: a finite path that satisfies $\delta_0, \ldots, \delta_N$, since a and b are $N+1$ steps apart. Bottom: a model of $\Gamma \cup \{\delta_n : n \geq 0\}$, in which a and b lie in different pieces, so it is not connected and cannot satisfy X.

***Exercise** 6.1.2* Write out δ_0 and δ_1, and check that the path in Figure 6.1 satisfies them when $N \geq 1$.

By Theorem 6.1.1, no set of sentences, and in particular no single sentence, defines connectivity. On finite graphs no single sentence defines connectivity either, but compactness cannot show this, since the models obtained from it may be infinite; the usual proof uses Ehrenfeucht–Fraïssé games and is beyond the course.[1]

***Exercise** 6.1.3* A graph is *bipartite* if its vertices can be split into two sets with every edge going between them; equivalently, it has no closed walk of odd length (a walk from a vertex back to itself). Show that the bipartite graphs are the models of a set of sentences, but not of a single sentence. (If φ worked, consider $\Gamma \cup \{\neg\varphi\}$ together with sentences saying there is no closed walk of odd length at most n.)

***Exercise** 6.1.4* Show that no set of sentences of L has as its models exactly the graphs in which every vertex has finitely many neighbours. (Add constants $c, d_1, d_2, \ldots$ with $E(c, d_i)$ and $\neg(d_i \approx d_j)$.)

In the next exercises, a *cycle of length* $n \geq 3$ in a graph is a sequence of distinct vertices $s_1, \ldots, s_n$ with s_i joined to s_{i+1} for $i < n$ and s_n joined to s_1. A graph is *acyclic* if it has no cycle, and a *tree* if it is connected and acyclic.

[1] R. Fraïssé, "Sur quelques classifications des systèmes de relations", *Publications Scientifiques de l'Université d'Alger, Série A* 1 (1954), 35–182; A. Ehrenfeucht, "An application of games to the completeness problem for formalized theories", *Fundamenta Mathematicae* 49 (1961), 129–141. In such a game two players compare two structures move by move; the second player has a winning strategy in the n-move game if and only if the structures satisfy the same sentences with at most n nested quantifiers.

***Exercise** 6.1.5* Let K be a class of graphs. Suppose that there are sets Y and Y' of sentences such that a graph is in K if and only if it satisfies Y, and is outside K if and only if it satisfies Y'. Show that some single sentence φ has the same property as Y. (The set $\Gamma \cup Y \cup Y'$ is not satisfiable.)

***Exercise** 6.1.6* Show that the acyclic graphs are the models of a set of sentences but not of a single sentence. Using Exercise 6.1.5, conclude that no set of sentences has as its models exactly the graphs that contain a cycle.

***Exercise** 6.1.7* Show that no set of sentences has as its models exactly the trees. (Adapt the proof of Theorem 6.1.1.)

6.2 *Nonstandard models of arithmetic*

Let L_A have a constant 0, a unary function symbol S (successor), binary function symbols + and ·, and a binary relation symbol <. The *standard model* $\mathbb{N}$ has the natural numbers as elements, with the usual meanings. For $n \geq 0$, the *numeral* $\underline{n}$ is the closed term $S(S(\cdots S(0)\cdots))$ with n copies of S; its value in $\mathbb{N}$ is n. Let $\mathrm{Th}(\mathbb{N})$ be the set of all sentences of L_A that are true in $\mathbb{N}$.

Theorem 6.2.1 (Skolem, 1934) *There is a structure M with $M \models \mathrm{Th}(\mathbb{N})$ and an element c^M greater than $\underline{n}^M$ for every n.*

Proof. Add a new constant c and let $X^+ = \mathrm{Th}(\mathbb{N}) \cup \{\underline{n} < c : n \geq 0\}$. A finite subset of X^+ mentions $\underline{n} < c$ only for $n \leq N$, for some N, and $\mathbb{N}$ with c read as $N + 1$ satisfies it. By Theorem 4.2.10, X^+ has a model.

So $\mathrm{Th}(\mathbb{N})$, the set of first-order sentences true in $\mathbb{N}$, does not determine $\mathbb{N}$: it has a model with an "infinite" element. By the Löwenheim–Skolem theorem we may take M countable. Such an M is a *nonstandard model of arithmetic*. Thoralf Skolem built the first one in 1934, without compactness.[2] Call the values $\underline{n}^M$ the *standard* elements of M.

The order of a nonstandard model Every sentence true in $\mathbb{N}$ holds in M. This gives the following description of $<$ in M.

- The values $\underline{n}^M$ form a copy of $\mathbb{N}$ at the start of M. The sentences $\underline{m} < \underline{n}$ for $m < n$, $\forall x\, \neg(x < \underline{0})$, and, for $n \geq 1$, $\forall x\, (x < \underline{n} \supset x \approx \underline{0} \vee \ldots \vee x \approx \underline{n-1})$ are true in $\mathbb{N}$. Hence the $\underline{n}^M$ are distinct and increasing, and nothing in M lies below $\underline{n}^M$ except $\underline{0}^M, \ldots, \underline{n-1}^M$.
- The other elements are *nonstandard*; since $<$ is a linear order in $\mathbb{N}$, it is one in M, so every nonstandard element is greater than every $\underline{n}^M$.

[2] T. Skolem, "Über die Nicht-charakterisierbarkeit der Zahlenreihe mittels endlich oder abzählbar unendlich vieler Aussagen mit ausschliesslich Zahlenvariablen", *Fundamenta Mathematicae* 23 (1934), 150–161. His elements were definable functions from $\mathbb{N}$ to $\mathbb{N}$, compared on a suitably chosen infinite set of arguments. Jerzy Łoś made this construction, the *ultrapower*, a general tool in 1955. Abraham Robinson later used nonstandard models of the real numbers to make infinitesimals rigorous.

- Every nonzero element of $\mathbb{N}$ has a predecessor, so every nonstandard d has a predecessor $d - 1$, which is again nonstandard. The sentences $\forall x\,(x < S(x))$ and $\forall x\,\forall y\,\neg(x < y \wedge y < S(x))$ are true in $\mathbb{N}$, so the elements $\ldots, d-1, d, d+1, \ldots$ form a copy of $\mathbb{Z}$, called the *block* of d.
- There is no first block and no last block, and between any two blocks there is a third (Exercise 6.2.3).

In summary, M is a copy of $\mathbb{N}$ followed by copies of $\mathbb{Z}$, with no first or last copy and a copy between any two (Figure 6.2).

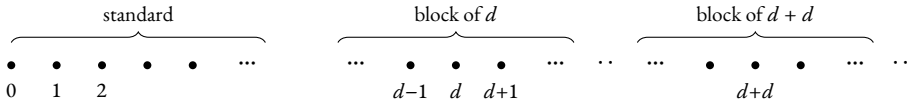


Figure 6.2 A nonstandard model of arithmetic: the standard numbers, then blocks that each look like $\mathbb{Z}$. Between any two blocks there is another, and there is no last block.

***Exercise** 6.2.2* Let d be nonstandard. Show that $S(d)$ is nonstandard and that $d < S(d)$, using sentences true in $\mathbb{N}$.

***Exercise** 6.2.3* Let $d < d'$ be nonstandard and in different blocks. Using sentences true in $\mathbb{N}$, show that $d + d$ is not in the block of d, that some nonstandard element lies below the block of d, and that some element lies strictly between the blocks of d and d'. (Every element of $\mathbb{N}$ is of the form $y + y$ or $y + y + 1$; halve d, and halve $d + d'$.)

***Exercise** 6.2.4* Show that no formula $\varphi(x)$ of L_A defines the standard elements, in the sense that $(M, \sigma[x \mapsto s]) \vDash \varphi(x)$ exactly when s is standard. (If it did, the sentence $(\varphi(0) \wedge \forall x\,(\varphi(x) \supset \varphi(S(x)))) \supset \forall x\,\varphi(x)$, which is true in $\mathbb{N}$, would hold in M.)